

OCCLUSION IS DISTRIBUTED IN VISION-LANGUAGE-ACTION POLICIES: SELECTIVE AND ATTRIBUTED TRANSCODER FEATURES FAIL HELD-OUT CAUSAL TESTS

Anonymous authors

Paper under double-blind review

ABSTRACT

Mechanistic claims about vision-language-action (VLA) policies usually select internal features by how selectively they respond to a concept and then show their causal role on the same inputs. We ask whether such claims survive a held-out, causally grounded protocol, for one concrete question: how does a VLA register that the object named in its instruction is hidden, and which internal features carry that into its actions? We train per-layer TopK transcoders on the Gemma backbone of $\pi_{0.5}$, hide the target with simulator-rendered occluders (an opaque box and a camera-blocking screen, plus a pixel-paint control) on all 10 LIBERO-Spatial tasks, fix every feature set on discovery demonstrations, and test on disjoint ones with task-clustered confidence intervals. Occlusion-selective features exist (452 of 73,728) and their selectivity replicates ($110\times$ the base rate), yet patching them closes 0.2% of the occlusion effect on the action, indistinguishable from norm-matched random features (0.0%). Even the top-100 features by gradient attribution close only 1.5 [0.6, 2.5]% (random: 0.0%), against 73.3% for all MLP outputs. The two sets barely overlap (8 shared features; Spearman $\rho=-0.02$ between selectivity and attribution). In closed loop the box lowers success from 97% to 17%; restoring the attributed features’ clean values at every inference step does not reliably recover it (-3 [-15 , $+5$] points). The pattern holds for $\pi_{0.5}$ on LIBERO-Object. Selectivity is not a reliable proxy for causal relevance in VLAs; held-out, attribution-based tests should be the default.

1 INTRODUCTION

Vision-language-action (VLA) models map camera images and an instruction to robot actions by fine-tuning a vision-language backbone with an action head (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024; Physical Intelligence et al., 2025). A growing literature opens them up with sparse autoencoders, probes and steering (Häon et al., 2025; Buurmeijer et al., 2026; Swann et al., 2026; Grant et al., 2026; Jin et al., 2026; Shi et al., 2026). A recurring pattern, here and in circuit analysis of language models (Marks et al., 2025; Ameisen et al., 2025), is to select features because they respond *selectively* to a concept and then to demonstrate their causal role on the same inputs. Selection and test on the same data inflate effects (Kriegeskorte et al., 2009), and selective responses need not be the ones the network uses (Bolukbasi et al., 2021; Makelov et al., 2024).

We study both risks on a question that matters for embodied agents: when the object named in the instruction is hidden, how does the policy register it, and which internal features carry that into the action? Answering it requires three things that earlier VLA analyses lack: occluders that remove the object from view (not just recolour its pixels), feature sets chosen on data disjoint from the test, and a causal ranking of features rather than a selectivity score. Our contributions:

- **A held-out, causally grounded circuit protocol for VLAs.** Feature sets are fixed on discovery demonstrations and tested on disjoint ones and on unseen tasks; frames sit at fixed phases of the grasp approach; every effect has a norm-matched random control and a task-clustered CI (§3).

- **Rendered occluders with matched controls.** An opaque box and a camera-blocking screen are drawn into the simulator’s renderer, with an identical-size occluder placed elsewhere and an occluder on a distractor object as controls (§3.1).
- **Selectivity versus attribution.** Neither selective nor attribution-selected sparse sets carry the effect, and we measure how far first-order attribution predicts patching (Pearson $r=0.28$; §4.3).
- **Closed-loop tests** that restore or inject feature values at every inference step over paired episodes (§4.4).
- **Replication** across a second suite and a second policy, and a within-study measurement of how in-sample selection inflates effects (§4.5).

2 RELATED WORK

Interpreting VLAs. Sparse autoencoders and probes have been used to find and steer interpretable directions in VLAs (Häon et al., 2025; Buurmeijer et al., 2026; Swann et al., 2026; Jin et al., 2026), to compare feature types (Grant et al., 2026) and to trace representations to behaviour (Shi et al., 2026; Zhang et al., 2026). These studies typically select features by activation statistics and test on overlapping data. **Transcoders and circuits.** Transcoders replace an MLP with a sparse, interpretable approximation that supports feature-level causal tracing (Dunefsky et al., 2024; Ameisen et al., 2025; Lindsey et al., 2025), including in VLMs (Damianos et al., 2026); we use TopK sparsity (Gao et al., 2025). **Attribution patching** approximates activation patching with one backward pass (Nanda, 2023; Syed et al., 2024; Kramár et al., 2024); we use it to rank transcoder features and measure where its first-order approximation holds. **Patching methodology.** We follow recommendations on metrics and controls (Zhang & Nanda, 2024; Heimersheim & Nanda, 2024) and add held-out selection and norm-matched random controls.

3 METHOD

Policy and tasks. $\pi_{0.5}$ (Physical Intelligence et al., 2025) (lerobot/pi05_iberofinetuned, LeRobot(Cadene et al., 2024))mapstwicameraimages, theinstructionandtherobotstatestepactionchunkwithaflow-matchingactionexpert(Lipman et al., 2023)thatreadsthekey/valuecacheofan18-layerGemmabackbone(GemmaTeam, 2024; Beyer et al., 2024).Weuseall10LIBERO – Spatialtasks(Liu et al., 2023)inMuJoCo/robosuite(Todorov et al., 2012; Zhu et al., 2020);thetargetistheobjectthein

Frames and split. For every task we replay demonstrations [0, 1] (*discovery*) and [2, 3, 4] (*test*) and take one frame at each of the fixed phases [0.3, 0.55, 0.8] of the approach to the target (nearest step where the target is visible and every occluder and control can be placed), giving 46 discovery and 70 test frames. No frame is chosen by the size of its effect. Transcoders are trained on discovery frames and 419 frames from further demonstrations; test frames never enter training or selection.

Metric. For a frame, a_b and a_o are the action chunks predicted from the base and occluded images with identical noise. A patch that turns a_b into a_p closes $100(1 - \|a_p - a_o\|/\|a_b - a_o\|)$ % of the gap (v1/v2 metric); we also report the projection $100\langle a_p - a_b, a_o - a_b \rangle / \|a_o - a_b\|^2$, which is linear in the patch. Frames with a gap below 0.1 are excluded (pre-specified).

3.1 OCCLUDERS AND CONTROLS

We draw occluders into MuJoCo’s scene at every render call (environment observations and segmentation alike); they have no physics and are recomputed from the live state, so they follow the target (Figure 1). **BOX:** an opaque box around the target’s bounding spheres, seen by both cameras. **SCREEN:** a thin panel on the line from the third-person camera to the target, sized to hide it from that camera only. **PAINT:** the v1/v2 edit that paints the target’s pixels gray. Each occluder has a *miss* control (the same occluder placed where it hides neither the target, nor the distractor, nor other task objects, found by a segmentation-checked search) and, for BOX and PAINT, a *distractor* control (the same occluder on the nearest object of the same type). *Recolour* and *absent* (target moved out of view) complete the conditions.



Figure 1: The conditions on one frame per task (third-person camera, as the policy sees it): base, recolor, absent, the three occluders (PAINT, BOX, SCREEN), their miss controls and the distractor controls.

3.2 TRANSCODERS, SELECTIVE FEATURES AND ATTRIBUTION

For every layer ℓ we train a TopK transcoder (Dunefsky et al., 2024; Gao et al., 2025) with 4096 features and $k=64$ that maps the MLP input to its output, $y_\ell \approx \sum_j f_j(x_\ell) d_j + b$. Setting a set S of features to their values in another run adds $\sum_{j \in S} (f'_j - f_j) d_j$ to the MLP output and keeps the transcoder’s error term.

Selective features (Goal 1) follow the v2 rule on discovery frames: a feature is selective for an occluder if its summed activation rises on $\geq 90\%$ of frames, by at least its typical activation, and by at least twice its largest response to recolor, absent and the miss control (specificity ≥ 0.5); the capped set keeps the 8 best per layer.

Attribution. For each frame we backpropagate the projection metric through the full sampler (all denoising steps) to an additive perturbation of every MLP output, and score feature j at layer ℓ by $\alpha_j = \sum_t (f_{j,t}^o - f_{j,t}^b) \langle d_j, \partial m / \partial y_{\ell,t} \rangle$ (Nanda, 2023; Syed et al., 2024), the first-order effect of setting it to its occluded value in the base run; the remove direction (occluded run, base values) is scored the same way and the two are averaged. The attribution set is the top-100 features by mean discovery-frame attribution among layers whose residual stream carries the occlusion signal (a single-layer residual swap closes $\geq 10.0\%$ of the gap on discovery frames: layers [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]).

Pre-specified tests. H1: selective features replicate on test frames. H2-A: the attribution set beats norm-matched random features by ≥ 10.0 points with a 95% CI above 0 (primary: BOX). H2-S: the v2 rule for the selective set. H3: selectivity and attribution dissociate ($< 10\%$ of the attribution set is selective and Spearman $\rho < 0.2$). H4: in closed loop, restoring the attribution set’s clean values under the occluder raises success relative to no restore and to a matched random set. CIs resample the 10 tasks with replacement; p -values are sign-flip tests over task means.

Table 1: Setup and validity battery. Checks: (1) edits change only the intended pixels, (2) determinism and bit-identical batched patching, (3) the occlusion effect exceeds the minimum gap, (4) a single-layer residual swap closes $\geq 80\%$ of it, (5) transcoders reconstruct ≤ 0.2 FVU. Gap: median action RMSE between base and occluded (primary occluder). FVU on held-out tokens / held-out test frames.

run	tasks	frames (disc.+test)	checks 1–5	gap	$\geq$ min	best swap (%)	FVU
$\pi_{0.5}$ / LIBERO-Spatial	10	46+70	✓✓✓✓✗	0.911	100%	94.4	0.204 / 0.294
$\pi_{0.5}$ / LIBERO-Object	10	60+60	✓✓✓✓✓	0.666	100%	94.4	0.183 / 0.334

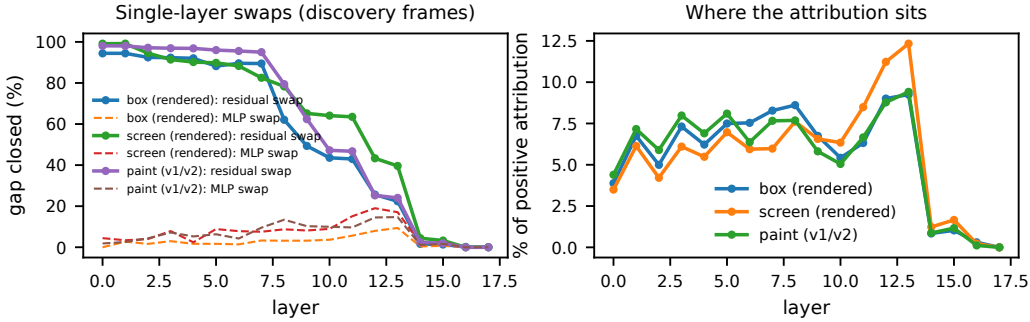


Figure 2: Left: single-layer swaps of the full residual stream (solid) or the MLP output (dashed) from the occluded into the base run, discovery frames. Right: where the positive attribution mass sits.

4 RESULTS

4.1 THE INSTRUMENT WORKS

All occluders change the action: the median base-vs-occluded gap is 0.911 (BOX), 0.366 (SCREEN) and 0.229 (PAINT), and occluding the target moves the action more than occluding the distractor (0.956 vs 0.161 for BOX). A single-layer residual swap closes up to 94.4% (layer 1); swaps after layer 13 close little (Figure 2). Transcoders reach a median FVU of 0.204 on held-out tokens and 0.294 on held-out test frames (v2: 0.185 and 0.400). Batched patching is bit-identical to single-row patching, so all patches of a frame share one forward pass. Table 1 lists every check.

4.2 Q1: SELECTIVE FEATURES EXIST AND REPLICATE

452 of 73,728 features pass the selectivity rule for BOX (417 also against the distractor control; top layers [6, 5, 8]). On held-out test frames 84% of the selected features pass the same rule again, against a base rate of 0.76% (110 $\times$), and discovery and test responses correlate (Spearman $\rho=0.82$). Selectivity is therefore a real, reproducible property of these features.

4.3 Q2: WHICH FEATURES CARRY THE EFFECT INTO THE ACTION?

Table 2 gives the held-out causal tests. The selective set closes 0.16 [−0.11, 0.50]% of the gap, no more than matched random features (+0.12 [−0.14, +0.45] points). The 100 attribution-selected features close 1.5 [0.6, 2.5]%, +1.5 [+0.6, +2.5] points above matched random features (H2-A fails), 2% of the MLP-output ceiling (73.3%). Chosen on the other half of the tasks only, they still close 0.7 [0.3, 1.0]%.

How many features? Figure 3 ranks features three ways. The attribution ranking reaches half of the MLP ceiling with more than the largest K tested features; ranking by activation change needs more than the largest K tested, and random features barely move the action. The oracle curve (attribution computed on the test frame itself) bounds what any ranking could achieve. All transcoder features together close 56.8%; the transcoders’ error term alone closes 6.9%.

Table 2: Held-out causal tests on $\pi_{0.5}$ / LIBERO-Spatial. Mean % of the base $\rightarrow$ occluded action gap closed by setting a feature set to its occluded values in the base run (inject), with 95% task-clustered bootstrap CIs. Random sets match the per-layer counts and per-token norm of the set they control for. Feature sets are fixed on discovery demonstrations.

	BOX	SCREEN	PAINT
attribution top- K	1.5 [0.6, 2.5]	2.8 [0.4, 5.0]	5.2 [3.3, 7.5]
random, matched	0.0 [−0.0, 0.1]	0.3 [−0.1, 0.9]	0.1 [−0.0, 0.3]
attribution, other tasks	0.7 [0.3, 1.0]	—	—
selective (Goal 1)	0.2 [−0.1, 0.5]	0.5 [0.0, 1.2]	−0.0 [−0.2, 0.2]
random, matched	0.0 [−0.0, 0.1]	−0.0 [−0.1, 0.1]	0.0 [−0.0, 0.1]
colour-selective	0.0 [−0.0, 0.0]	—	—
all transcoder features	56.8 [49.4, 67.8]	55.1 [51.3, 58.4]	59.0 [56.7, 61.2]
transcoder error only	6.9 [2.2, 12.0]	—	—
all MLP outputs	73.3 [67.5, 80.9]	78.5 [73.3, 83.1]	75.8 [70.2, 81.0]
attribution — random	+1.5 [+0.6, +2.5]	+2.5 [+0.2, +4.8]	+5.1 [+3.2, +7.3]
selective — random	+0.1 [−0.1, +0.5]	+0.5 [+0.1, +1.1]	−0.1 [−0.3, +0.1]
attribution — selective	+1.3 [+0.3, +2.5]	+2.4 [+0.2, +4.1]	+5.3 [+3.2, +7.6]
remove: attribution — random	+12.8 [+3.4, +22.5]	+6.8 [+4.2, +10.5]	+18.2 [+11.3, +26.0]
H2-A (attribution)	✗	✗	✗
H2-S (selective)	✗	✗	✗
held-out frames (tasks)	70 (8)	69 (8)	63 (8)

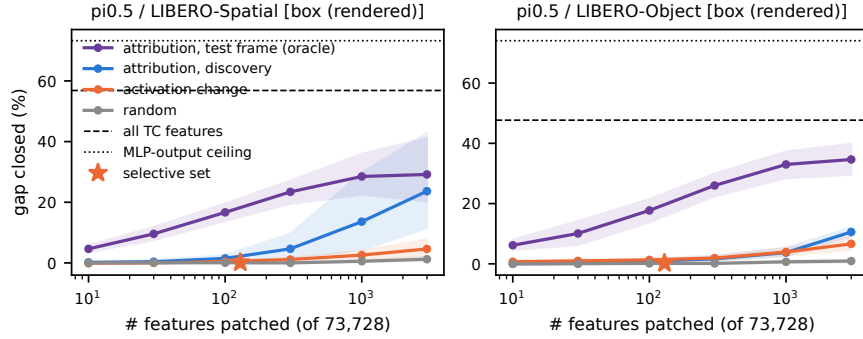


Figure 3: Gap closed on held-out frames by the top- K features of each ranking (shaded: 95% CI). Star: the selective set. Lines: all transcoder features and all MLP outputs.

Is attribution trustworthy? Across every set we patched, the attribution prediction and the measured projection correlate with Pearson $r=0.28$ (slope 0.06, $n=2870$; Figure 4), so first-order attribution is only a rough proxy here, which is why every claim rests on actual patches.

Selectivity and attribution dissociate. The attribution set and the selective set share 8 features; 8% of the attribution set passes the selectivity rule, selective features sit at the 59th attribution percentile (median) and carry 3.07% of the positive attribution, and across all features Spearman $\rho(\text{selectivity}, \text{attribution})=-0.02$ (Figure 4). H3 holds.

Where. Restricting the patch to tokens under the occluder closes 1.1 [0.6, 1.8]% (attribution set) and 45.6 [38.5, 56.8]% (all MLP outputs); the remaining tokens close 0.3 [0.0, 0.5]% and 9.9 [4.9, 15.3]%.

4.4 Q3: DOES IT MATTER IN CLOSED LOOP?

Table 3 runs 5 tasks $\times$ 12 paired episodes per condition. The box lowers success from 97% to 17% (+80 [+60, +97] points). Restoring all MLP outputs from a clean render at every inference step gives 98%, restoring the attribution set 13% and a matched random set 13% (restore attribution — random: +0 [−12, +13] points). Injecting the attribution set’s occluded values into clean episodes gives 100%. H4 does not hold.

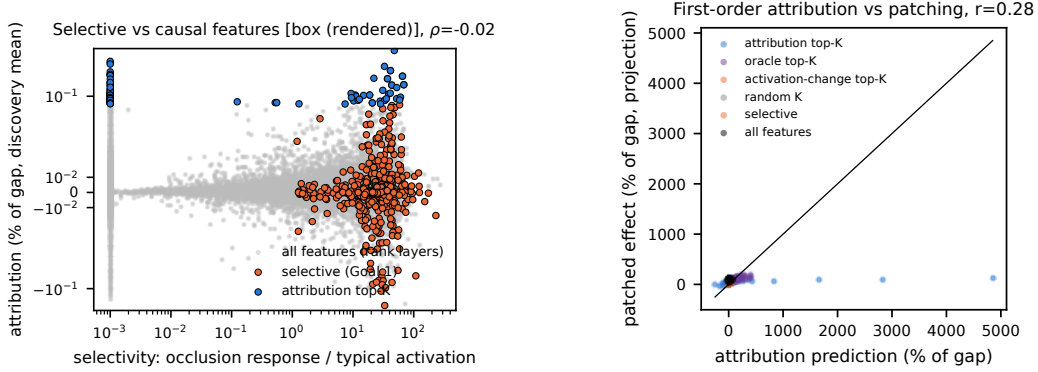


Figure 4: Left: selectivity vs attribution for every feature in the rank layers; orange: selective, blue: attribution set. Right: attribution prediction vs measured effect of every patched set.

Table 3: Closed loop on 5 tasks $\times$ 12 paired episodes (same initial states and noise seeds in every condition). Success with Wilson 95% CIs; *distractor*: episodes in which the other object was lifted. Restores replace the features’ values in the occluded run with their values from a clean render of the same state, at every inference step.

condition	success (%)	distractor (%)
clean	97 [89, 99]	0
box	17 [9, 28]	12
screen	55 [42, 67]	0
paint	73 [61, 83]	8
box + restore all MLP outputs	98 [91, 100]	0
restore attribution set	13 [7, 24]	10
restore random set (matched)	13 [7, 24]	12
clean + inject attribution set	100 [94, 100]	0
restore selective set	18 [11, 30]	12
paired difference (pts)	[95% CI]	McNemar p
clean — box	+80 [+60, +97]	0.000
restore attribution — no restore	−3 [−15, +5]	0.791
restore attribution — restore random	+0 [−12, +13]	1.000
restore all MLP — no restore	+82 [+68, +93]	0.000
clean — inject attribution	−3 [−10, +0]	0.500

4.5 GENERALIZATION AND THE COST OF IN-SAMPLE SELECTION

Table 4 repeats the core protocol on $\pi_{0.5}$ / LIBERO-Object. H2-A fails for $\pi_{0.5}$ / LIBERO-Object.

Measured inside one study, in-sample evaluation inflates effects: the selective set beats its matched random control by 0.22 points on the discovery frames it was chosen on and by 0.12 points on held-out frames; the attribution set beats its matched random control by 5.17 points on the discovery frames it was chosen on and by 1.45 points on held-out frames. Choosing frames the v1 way (the four largest-gap frames of one task) turns the selective set’s ratio to random from $3.9\times$ on all frames into $-4.1\times$. Table 5 places the three protocols side by side.

5 DISCUSSION AND LIMITATIONS

The occlusion effect on the action is carried by many features at once: neither the most selective features nor the top attribution-ranked ones account for a large share of it. Three limitations bound these claims. (i) The rendered occluders have no physics and follow the target, so their position still reveals where the target is; they remove its appearance, not its location. (ii) Attribution is first-order at the clean run; we therefore report its measured agreement with patching and base every claim on patches. (iii) Transcoders leave an error term whose share of the effect we report; features outside

Table 4: Replication across policies and suites (box occluder, held-out frames, % gap closed [95% CI]). Last column: H2-A, H2-S.

run	attribution	random	selective	random	all TC	MLP	H2
$\pi_{0.5}$ / LIBERO-Spatial	1.5 [0.6, 2.5]	0.0 [−0.0, 0.1]	0.2 [−0.1, 0.5]	0.0 [−0.0, 0.1]	56.8	73.3	XX
$\pi_{0.5}$ / LIBERO-Object	0.9 [0.5, 1.4]	0.0 [−0.0, 0.1]	0.2 [0.0, 0.4]	0.1 [0.0, 0.2]	47.7	74.1	XX

Table 5: The same question under three protocols. v1: in-sample pilot; v2: held-out selectivity study; v3: this paper.

	v1	v2	v3
tasks / probe frames	1 / 4 (max effect)	10 / 60 (fixed phases)	10 / 116 (fixed phases) + 2 replications
selection vs. test	same frames	disjoint demos	disjoint demos + cross-task
occluder	paint	paint	rendered box, screen + paint
transcoder	512, $k=16$	2,048, $k=32$	4096, $k=64$
FVU held-out / test frames	0.174 / –	0.185 / 0.400	0.204 / 0.294
features chosen by	selectivity	selectivity	selectivity and attribution
selective set, paint (%)	3.92	0.27 [−0.32, 0.92]	−0.01 [−0.21, 0.15]
random matched, paint (%)	0.56	0.05	0.05 [−0.00, 0.10]
attribution set, box (%)	–	–	1.5 [0.6, 2.5]
all TC / MLP ceiling (%)	40.1 / 65.5	42.8 / 76.6	56.8 / 73.3

the dictionary are invisible to both rankings. The unit of replication is the task; two suites and two policies bound but do not settle generality across embodiments.

6 CONCLUSION

Held-out selection, rendered occluders with matched controls and attribution-based rankings change the answer to a simple mechanistic question about a VLA. We recommend that VLA interpretability studies fix feature sets on data disjoint from the test, rank features by their effect on the action rather than by selectivity, and report norm-matched random controls with task-level uncertainty.

REFERENCES

- Emmanuel Ameisen, Jack Lindsey, Adam Pearce, et al. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- Lucas Beyer, Andreas Steiner, André Susano Pinto, et al. PaliGemma: A versatile 3B VLM for transfer. *arXiv preprint arXiv:2407.07726*, 2024.
- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.
- Tolga Bolukbasi, Adam Pearce, Ann Yuan, Andy Coenen, Emily Reif, Fernanda Viégas, and Martin Wattenberg. An interpretability illusion for BERT. *arXiv preprint arXiv:2104.07143*, 2021.
- Anthony Brohan, Noah Brown, Justice Carbajal, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.
- Hugo Buurmeijer, Carmen Amo Alonso, Aiden Swann, and Marco Pavone. Observing and controlling features in vision-language-action models. *arXiv preprint arXiv:2603.05487*, 2026.
- Remi Cadene, Simon Alibert, Alexander Soare, Quentin Gallouedec, Adil Zouitine, and Thomas Wolf. LeRobot: State-of-the-art machine learning for real-world robotics in PyTorch. <https://github.com/huggingface/lerobot>, 2024.
- Dimitrios Damianos, Leon Voukoutis, Georgios Skyrianos, Vassilis Katsouros, and Georgios Paraskevopoulos. Transcoders trace visual grounding and hallucinations in vision-language models. *arXiv preprint arXiv:2605.22902*, 2026.

- Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders find interpretable LLM feature circuits. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.11944.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025. arXiv:2406.04093.
- Gemma Team. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.
- Bryce Grant, Xijia Zhao, and Peng Wang. Not all features are created equal: A mechanistic study of vision-language-action models. *arXiv preprint arXiv:2603.19233*, 2026.
- Bear Häon, Kaylene Stocking, Ian Chuang, and Claire Tomlin. Mechanistic interpretability for steering vision-language-action models. In *Conference on Robot Learning*, 2025. arXiv:2509.00328.
- Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.
- Xinchen Jin, Aditya Chatterjee, Pranav Kumar, and Rohan Paleja. Event-grounded sparse autoencoders for vision-language-action policies. *arXiv preprint arXiv:2605.17204*, 2026.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
- János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. AtP*: An efficient and scalable method for localizing LLM behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024.
- Nikolaus Kriegeskorte, W. Kyle Simmons, Patrick S. F. Bellgowan, and Chris I. Baker. Circular analysis in systems neuroscience: the dangers of double dipping. *Nature Neuroscience*, 12(5): 535–540, 2009.
- Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, et al. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023. arXiv:2306.03310.
- Aleksandar Makelov, Georg Lange, and Neel Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024.
- Samuel Marks, Can Rager, Eric J. Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. In *International Conference on Learning Representations*, 2025. arXiv:2403.19647.
- Neel Nanda. Attribution patching: Activation patching at industrial scale. <https://www.neelnanda.io/mechanistic-interpretability/attribution-patching>, 2023.
- Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025.
- Haoyuan Shi, Xiancong Ren, Yingji Zhang, Qinfan Zhang, Jiayu Hu, Haozhe Shan, Han Dong, Jinpeng Lu, Yinda Chen, Yi Zhang, Yong Dai, and Xiaozhu Ju. VLA-Trace: Diagnosing vision-language-action models through representation and behavior tracing. *arXiv preprint arXiv:2605.30117*, 2026.

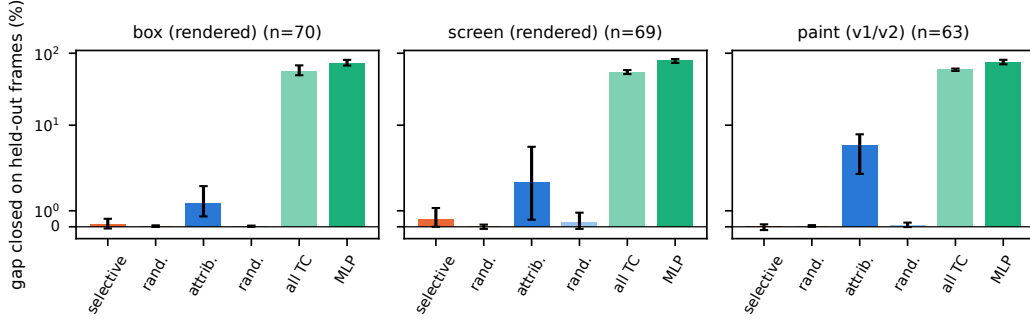


Figure 5: Held-out gap closed per occluder family and feature set (95% CIs).

Aiden Swann, Lachlain McGranahan, Hugo Buurmeijer, Monroe Kennedy III, and Mac Schwager. Sparse autoencoders reveal interpretable and steerable features in VLA models. *arXiv preprint arXiv:2603.19183*, 2026.

Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. In *Proceedings of the 7th BlackboxNLP Workshop*, 2024. arXiv:2310.10348.

Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012.

Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*, 2024. arXiv:2309.16042.

Hanxin Zhang, Mingshuo Xu, Abdulqader Dhafer, Shigang Yue, Hongbiao Dong, and Zhou Daniel Hao. Embodied interpretability: Linking causal understanding to generalization in vision-language-action models. *arXiv preprint arXiv:2605.00321*, 2026.

Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. *arXiv preprint arXiv:2009.12293*, 2020.

A ADDITIONAL RESULTS

Table 6: Gap closed (%) by the top- K features of each ranking, held-out frames, primary occluder.

ranking	$K=10$	$K=30$	$K=100$	$K=300$	$K=1000$	$K=3000$
attribution (discovery)	0.2	0.4	1.5	4.6	13.6	23.7
attribution (oracle)	4.6	9.6	16.7	23.4	28.5	29.2
activation change	-0.1	0.0	0.5	1.1	2.6	4.6
random	0.0	0.0	0.0	0.0	0.6	1.2

B CONFIGURATION

All settings of the main run (`config.json`): `attr_batch=3`, `attr_k=100`, `batch_invariance_tol=1e-05`, `capture_batch=8`, `capture_max_ram_frac=0.45`, `ceiling_frames=8`, `ceiling_min_pct=80.0`, `check3_min_frac=0.9`, `check5_blocks=False`, `clean_edit_max_outside=0.002`, `cover_margin=1.15`, `determinism_frames=8`, `determinism_tol=0.0`, `discovery_demos=[0, 1]`, `distractor_object=auto`, `extra_conds=['base', 'covered', 'screened', 'occluded']`, `extra_demos=[5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16]`, `extra_frames_per_demo=3`, `families=['cover', 'screen', 'paint']`, `force_continue=True`, `frame_phases=[0.3, 0.55, 0.8]`, `frame_search=6`, `frame_source=auto`, `goal1_batch=12`, `goal2_insamle=True`, `goal2_margin_pts=10.0`, `goal2_min_gap=0.1`, `goal3_episodes=12`, `goal3_init_offset=0`, `goal3_max_steps=0`, `goal3_occluder=auto`, `goal3_tasks=[0, 2, 6, 7]`,

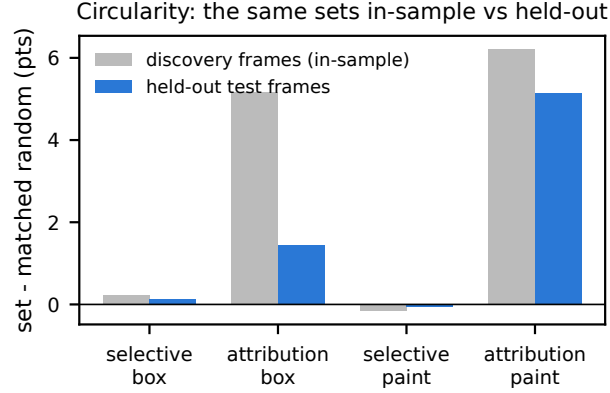


Figure 6: The same feature sets evaluated on the discovery frames they were chosen on and on held-out frames.

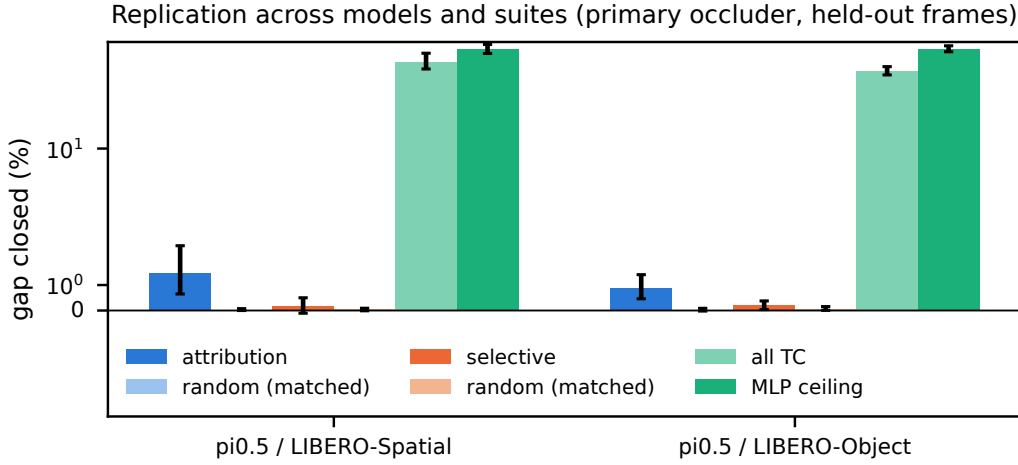


Figure 7: Replication across policies and suites.

9], kcurve_k=[10, 30, 100, 300, 1000, 3000], kcurve_k_remove=[30, 100, 300, 1000], lazy_cameras=True, lift_detect_m=0.05, lift_dz=0.01, max_features_per_layer=8, max_train_tokens=450000, min_gap=0.02, min_mask_px=80, min_pos_frac=0.9, min_rise_tokens=1.0, min_specificity=0.5, min_t=5, n_bootstrap=10000, n_permutations=200000, n_random_draws=8, n_random_draws_remove=3, n_random_draws_secondary=3, n_single_features=3, obs_size=256, occ_pad_px=2, occluder_rgba=[0.5, 0.5, 0.5, 1.0], paint_gray=128, patch_batch=8, policy_path=lerobot/pi05_libero_finetuned, primary_family=cover, rank_layers_fallback=0-13, rank_min_resid_pct=10.0, recolor_rgb=[200, 40, 40], render_max_outside=0.03, render_pad_px=3, save_videos=True, screen_camera=agentview, screen_frac=0.35, screen_scale=1.35, seed=0, shadow_pad_px=6, splice_frames=10, suite=libero_spatial, target_keyword=bow1, target_object=auto, task_id=0, task_ids=[0, 1, 2, 3, 4, 5, 6, 7, 8, 9], tc_aux_coef=0.03125, tc_aux_k=256, tc_batch=4096, tc_dead_steps=100, tc_features=4096, tc_holdout_frac=0.05, tc_k=64, tc_layers=all, tc_lr=0.002, tc_max_fvu=0.2, tc_retry_mult=1.5, tc_steps=8000, tc_tf32=True, test_demos=[2, 3, 4], token_min_frac=0.1, top_layers=3.

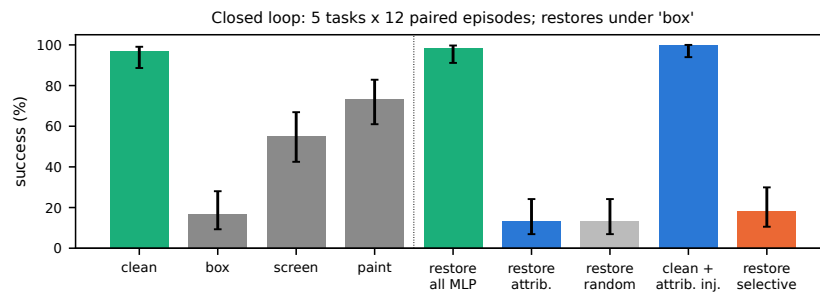


Figure 8: Closed-loop success with Wilson 95% CIs.